

Daniel J. Freed
William L. Martens

Northwestern Computer Music (NCM)
School of Music
Northwestern University
Evanston, IL 60201 USA
(312) 491-3895
ihnp4!numusic!dan
ihnp4!numusic!bill

ABSTRACT

A number of important methodological concerns for psychophysical research on timbre are discussed, including details of stimulus selection, perceptual measurement, acoustical measurement, and derivation of psychophysical relations. A pilot study is described, in which a set of recorded percussive sounds was subjected to acoustical analysis and perceptual rating. The result of the study is a relation which allows judgments of *perceived hardness of mallet* to be quantitatively predicted from acoustical parameters.

1. TIMBRE AND THE COMPUTER

Timbre has long been a favorite topic in computer music. The computer is an infinitely flexible sound-producing machine which presents unique possibilities for the invention and precise control of timbres. With this flexibility, however, comes complexity; the musician must be technically sophisticated in order to take full advantage of the possibilities. Fortunately, the computer may also be used as a high-precision acoustical analysis tool for psychophysical research on timbre. The results of such research are guidelines for musicians about how to achieve desired timbral effects with the computer.

When working with timbre, the musician must answer two basic questions:

1. "What do I want?"

The musician must decide what timbres to use and plan how to use them.

2. "How do I get it?"

The musician must discover how to coax the desired timbres out of the computer.

The first question is of a musical and psychological nature. Recent discussions in the literature are given by Martens (1985), who describes an environment for developing a "palette" of timbres, and McAdams and Saariaho (1985), who consider the organization of a collection of timbres into a musically functional structure. The current research focuses on the second question, which is of a psychophysical nature. The problem is to create a sound whose timbre matches the musician's perceptual description. In other words, what is wanted is *perceptual control over timbre synthesis*.

We seek knowledge about controlling timbre synthesis that is both *quantitative* and *universal*. By trial and error, musical practitioners may assemble "rules of thumb" that are effective for controlling a particular synthesis or processing algorithm (or hardware device), but such rules are generally only qualitative and not applicable to other synthesis methods. Developing basic understanding of timbre synthesis requires systematic, quantitative research. Furthermore, if research is to produce results that are applicable to any synthesis algorithm, it must focus on *acoustical parameters* rather than *synthesis parameters*. This approach has a disadvantage: additional knowledge is needed about the relation between acoustical and synthesis parameters in order to put the research results into practice. An important advantage, aside from increased generality, is that descriptions of timbre in terms of acoustical parameters more accurately represent the information presented to the auditory system. The ear is concerned only with the acoustical information present in a sound, and not with what knob settings or numerical values were used to produce the sound. It may be said that acoustical parameters are one step removed from practice, while synthesis parameters are one step removed from perception.

In summary, our task is to derive a collection of quantitative relations between the acoustical signal and the timbral percept it evokes. The problem is an old one, and the literature provides a number of possible research strategies to be considered (a more extensive literature review is given by Grey [1975]).

2. RESEARCH METHODOLOGY AND PHILOSOPHY

Any program of psychophysical research must include the following activities:

- Stimulus selection
- Perceptual measurement
- Acoustical measurement
- Derivation of psychophysical relations

We describe each of these steps in more detail below. In our discussions, we borrow a number of concepts from the school of *ecological psychology*, as represented by such investigators as Gibson (1966) and Neisser (1976), which are relevant to the study of complex percepts such as timbre.

2.1 Stimulus selection

It is common practice in psychoacoustical research to use stimuli which are acoustically and mathematically "simple," such as sine waves, square waves, and clicks. Because these stimuli can be completely described by a small number of parameters, they are attractive to researchers who seek to design carefully controlled experiments. However, the common wisdom among computer musicians is that such stimuli tend to have "electronic-sounding" timbres – flat, lifeless, unpleasant, and of little musical value – and that the synthesis of "natural-sounding" timbres requires acoustical complexity. Temporal variation of acoustical parameters has been shown to be a particularly important feature of natural sounds (Risset & Mathews 1969).

It seems intuitively obvious that musically relevant timbre research cannot employ musically useless timbres as stimuli. This intuitive reaction is supported by the ecological psychologists, who make a more general criticism about the use of highly simplified stimuli in perceptual research. Gibson (1966), Neisser (1976), and others argue that perceptual systems require complexity in order to operate normally. Natural stimuli contain a large number of parameters which vary in a *coherent, redundant* way. This excess of information, far from confusing the system, evokes stable and immediate percepts, because the coherence and redundancy resolves ambiguity. A well-known example in computer music is a phenomenon described by Chowning (1980), McNabb (1981), and McAdams (1982): when vibrato is added to an unconvincing synthetic vocal timbre, the coherent variations in frequency aid spectral fusion, and the realism of the timbre increases dramatically. Research results obtained using oversimplified stimuli have little relevance to perception under natural conditions – they lack what Neisser (1976) calls *ecological validity*.

It can be concluded that stimuli for timbre research should be acoustically complex and natural-sounding. The same requirements may be made of musically useful timbres. McAdams and Saariello (1985), however, caution against misinterpreting either of these requirements in musical applications, and their arguments apply to selecting stimuli for research as well. First, acoustical complexity alone is insufficient; it is *coherence* amidst the complexity that the auditory system seeks. Second, there is no need to restrict the choice of timbres to include only natural sounds or sounds synthesized from a physical model. Unusual, unnatural, even bizarre timbres are suitable for music and research, as long as they provide the coherent complexity necessary to engage the auditory system.

In our view, failure to consider these issues when selecting stimuli is a serious flaw in many of the psychophysical studies of timbre.

2.2 Perceptual measurement

Measuring the perception of timbre requires collecting subjective judgments of stimuli from subjects. Timbre,

however, is *multidimensional* – it is comprised of many distinct components (or *dimensions*). This presents a difficulty: what should subjects be instructed to judge? Certain dimensions of timbre, such as brightness, seem obvious enough that some researchers have simply chosen a likely timbral dimension to investigate. There are, however, more systematic approaches to the problem.

One common approach is to require subjects to rate each stimulus on a large number of semantic scales (such as "bright/dark", "full/empty", etc.). Principal components analysis (or similar techniques, such as factor analysis or dimensional analysis) is then used to measure correlations between different scales and determine a small set of uncorrelated scales (*principal components*) which represent all the data (Solomon 1958; von Bismarck 1974a). Grey raises an important objection to this procedure: "words may not exist to describe certain perceived differences, and even if such words do exist, they may not have been included among the scales available to the subjects" (1975, pp. 7–8). A related problem is that subjects may not agree on the definition of a word like "full" as it applies to timbre.

Semantic problems may be avoided through *multidimensional scaling* (MDS). In this procedure, subjects rate the similarity of pairs of timbres; these data are used to construct a space in which each timbre is represented by a point. Each dimension of the space may be interpreted as a dimension of timbre. Since subjects rate similarity rather than position on a verbal scale, semantic difficulties are minimized. A further advantage is that similarity judgments are *synthetic* rather than *analytic* (the terminology follows the example of McAdams 1982); they require the subject to consider the timbre as a whole, rather than attending selectively to a single dimension. We consider synthetic listening to be a more natural task than analytic listening.

If MDS is to be used for determining dimensions of timbre, proper stimulus selection is crucial. Some investigators (Plomp & Steeneken 1969; Miller & Carterette 1975; Howard & Silverman 1976; de Bruijn 1978) construct a synthetic stimulus set in which only a small number of acoustical parameters are allowed to vary. This strategy is useful for determining the relative saliency of the chosen acoustical parameters; for example, Plomp & Steeneken (1969) were able to measure the importance of phase relative to harmonic amplitude pattern. However, in order to discover the natural dimensions of timbre, the stimulus set must contain the wide range of acoustical variation typical of a collection of natural stimuli (Plomp 1970; Wedin & Goude 1972; Grey 1975, 1977; Wessel 1979; see also Cermak & Cornillon 1976, who apply the technique to traffic noise).

The main shortcoming of MDS is that it does not *identify* the dimensions of timbre. The MDS solution space indicates the structure of timbre space, but it is left to the researcher to conjecture as to what perceptual property is represented by each axis of the space. Such conjectures can only be tested by collecting additional data in which subjects rate timbres analytically with respect to the proposed perceptual property.

We chose to collect direct ratings of perceptual properties rather than similarity judgments, in order to avoid the need to conjecture about the identity of perceptual dimensions. However, we required our perceptual scales to be *ecologically relevant*. According to Gibson's (1966) theory of *direct perception*, the function of perceptual systems is to collect practical, usable information about the environment, and therefore the domain of perception is *objects* rather than *sensations*. Ecologically relevant scales for timbre, therefore, should require the subject to make a judgment about *perceived physical properties* of the sound source, such as perceived size, rather than abstract judgments about sensory qualities, such as "brightness" or "fullness." VanDerveer (1979) has found that listeners tend to describe sounds in terms of how they might have been produced; only in the pathological case are sounds described *introspectively*, in terms of the sensations they evoke, rather than *ecologically*, with reference to the environment. Rating perceived physical properties should be a natural task which produces data representative of the normal operation of the auditory system.

Direct scaling of perceived physical properties retains two important advantages of the MDS method. First, these scales do not suffer from the semantic ambiguity of abstract scales. Second, judgments on these scales require synthetic listening. If a variety of dissimilar acoustical cues are all consequences of the same physical property of the sound source, they will be combined by the auditory system into a single percept. Studdert-Kennedy cites an example from speech perception: "perceived place of articulation of an English stop consonant /p, t, k/, induced by a brief silence between /s/ and a following vowel..., depends on the length of the silence, on spectral properties at the offset of /s/, and on the relation between those properties and those of the following vowel" (1985, p. 144). All these cues are consequences of a single articulatory variable - voice onset time - and thus are integrated by the system into a unified percept.

Collecting ratings of perceived physical properties of stimuli does not obligate the researcher to measure *actual* physical properties of the sound source. It is true that the auditory system attempts to make judgments about how the sound was produced, but the only information available to the system is the acoustical signal. The ear is unable to distinguish between an actual gunshot and a (sufficiently faithful) recording of a gunshot, since the acoustical information is identical in both cases. To control timbre synthesis it is only necessary to understand how perception depends on the acoustical signal. This is especially true in computer music, where synthesized rather than mechanically-produced sounds are the norm.

While ecologically relevant scales have been utilized in vision research, they are rare in auditory research (an exception is Warren & Verbrugge 1984). We feel that the use of such scales holds considerable promise for future psychophysical research on timbre.

2.3 Acoustical measurement

The perceptual measurement question presented by the multidimensionality of timbre has its parallel in the domain of acoustical measurement. From among all the measurable acoustical parameters, a few must be chosen. We seek acoustical parameters that describe the signal in a succinct, perceptually relevant way.

2.3.1 Data reduction. The description of an acoustical signal with a large number of *local* parameters (such as time-varying frequency-band energies) is easily obtained. For deriving psychophysical relations to use in the eventual synthesis of timbres from perceptual descriptions, however, a manageably small set of data is required; *global* parameters are needed which capture general trends rather than details. A global description is incomplete in the sense that it is impossible to exactly resynthesize the original sound from the description. However, if the global parameters successfully capture the most important acoustical properties of the sound, the loss of local detail should be perceptually unimportant.

A common approach in the literature is to perform short-time Fourier analysis to obtain amplitude and frequency traces in each of a number of spectral bands. This data is then reduced by fitting curves to the traces (Grey 1975, 1977; Charbonneau 1981) or by performing principal components analysis (or a similar technique) to determine correlations among the traces (Plomp, Pols, & van de Geer 1967; Wedin & Goude 1972; Zahorian & Rothenberg 1981; Searle 1982). Resynthesis from the reduced data is performed to check the perceptual effects of the reduction (Grey & Moorer 1977; Charbonneau 1981; Zahorian & Rothenberg 1981).

Principal components analysis may also be used to choose from among a set of related global parameters. Cernak & Cornillon (1976) measured 14 parameters of recordings of traffic noise which they subjected to principal components analysis. They found the parameters to be highly redundant, and they were able to sufficiently capture the information contained in all 14 parameters using only the 2 parameters which had the highest correlation with the principal components.

The technique of principal components analysis depends on exploiting redundancy among acoustical parameters. It is useful to distinguish three different classes of redundancy:

1. Tautological redundancy.

The parameters are correlated because they measure the same thing.

2. Ecological redundancy.

The parameters are always correlated in natural sounds, but artificial stimuli can be synthesized in which the parameters vary independently. These artificial stimuli are likely to produce unusual percepts.

3. Identifying redundancy.

The correlation between the parameters is an identifying characteristic of the class of timbres under investigation.

When performing data reduction by eliminating redundancy, it is important to consider these distinctions. For example, if the redundancy is an identifying feature of some class of timbres, then the data reduction scheme is not applicable outside that class of timbres, whereas if the redundancy is tautological, the data reduction may be applied universally.

2.3.2 Perceptual relevance. The surest way to verify that an acoustical parameter is perceptually relevant is to measure its correlation with a perceptual measure. This is especially useful for choosing the best predictor of perception from among a number of related acoustical parameters (Cermak & Cornillion 1976; Grey & Gordon 1978). It is convenient, however, to have some *a priori* criteria for devising acoustical measures.

The principles of direct perception (Gibson 1966) suggest that the most appropriate acoustical parameters are *physical correlates* – measures which carry information about physical properties of the sound source. Physical correlates may be determined theoretically or empirically. The theoretical approach is to design an acoustical model, of which there are many examples in the literature (acoustical modeling for musical purposes in particular is discussed by Hiller & Ruiz 1971; Cadoz, Luciani, & Florens 1984; Adrien & Rodet 1985). The empirical approach is to measure various parameters of natural sounds and to observe how they relate to physical properties. Some parameters may be found to vary systematically with changes in the sound-producing system (Fletcher & Bassett 1978; Rossing & Sykes 1982); conversely, it may be possible to identify parameters which remain constant across a set of physically similar sound sources, such as an instrument family (Luce & Clark 1967).

The auditory system's affinity for acoustical complexity (see Sec. 2.1) is, according to Gibson, a consequence of the acoustical complexity of physical correlates. "...mathematical complexities... [are the] simplicities of auditory information... [and are] distinguished naturally by the auditory system. Moreover, it is just these variables that are specific to the source of the sound" (1966, p. 87). Gibson distinguishes between *low-order* properties – standard acoustical parameters such as frequency, amplitude, rate of decay, etc. – and *high-order* properties derived from low-order ones, such as rates of change of low-order properties, correlations between several low-order properties, or deviations of low-order properties from an average value or trajectory. Traditional psychoacoustics assumes that the auditory system is only sensitive to low-order properties, and operates by assembling bits of individually meaningless sensory data into the perception of a sound object. Gibson's alternative hypothesis is that high-order properties which are directly related to properties of the sound source are perceived directly, and no "assembly" process is required.

2.4 Derivation of psychophysical relations

Although there have been many studies of the perceptual components and physical correlates of timbre, few investigators have taken the next step of quantifying the relation between the two. One approach has been to relate similarity judgments to a measure of acoustical similarity. Plomp (1970) used a single-valued measure of spectral similarity for this purpose. Wedin & Goude (1972) also measured spectral similarity, but they expressed it as a vector and applied multiple regression to derive a psychophysical relation. Another approach has been to search for a single acoustical parameter which correlates with an axis of an MDS solution space (Wedin & Goude 1972; Grey & Gordon 1978; Wessel 1979). Factor analysis may be helpful to determine parameters for this purpose (Wedin & Goude 1972). Other investigators have measured a variety of acoustical parameters and tested the effect of each on direct judgments of timbral qualities, such as sharpness (von Bismarck 1974b) or roughness (Terhardt 1974).

As mentioned above (Sec. 2.2), we chose to collect direct judgments of a single perceptual property rather than similarity judgments. However, even when only a single timbral dimension is being investigated, it is important to measure more than one acoustical parameter. Perceptual judgments – especially judgments about perceived physical properties – depend on a variety of acoustical cues (see Sec. 2.2). We therefore chose to follow the example of von Bismarck (1974b) and Terhardt (1974) by considering each perceptual judgment to be a function of a collection of physical variables. We used multiple regression techniques to derive a many-to-one psychophysical relation.

3. AN EXPERIMENT WITH SOME PERCUSSIVE TIMBRES

We performed a pilot study to explore the perception of a set of related timbres. In designing this study we made the following decisions:

- We performed both acoustical and perceptual measurement in order to derive quantitative psychophysical relations.
- We used recordings rather than synthesized sounds as stimuli to provide natural acoustical complexity and coherence.
- We chose to work with percussive sounds, for two reasons: first, percussive sounds have received considerably less attention in the literature than nonpercussive sounds; second, percussive sounds are prevalent in the environment, and can be used to make ecologically relevant judgments.
- We collected direct scaling judgments rather than similarity judgments to avoid uncertainty in identifying perceptual dimensions.
- We chose to scale a perceived physical property for the sake of ecological relevance.

- As acoustical parameters, we used high-order, global measures that were empirically determined to be correlated with physical properties of the sound source, because we expected these parameters to be perceptually relevant.
- We also included acoustical parameters that we did *not* expect to be relevant to test whether our method for deriving relations would distinguish between relevant and irrelevant parameters.
- We chose to derive a many-to-one relation expressing a perceptual result as a function of several acoustical parameters, on the theory that most percepts are determined by more than one acoustical feature.

3.1 Stimuli

For our stimuli, we made recordings of several pieces of metal cookware being struck with a variety of percussion mallets. A number physical variables are represented by this set, including the size, shape, weight, and material of the metal pan, and the size, weight, and hardness of the mallet. All possible combinations of pans and mallets were recorded, so that we could investigate the listener's ability to make mallet-related judgments regardless of which pan was struck, or pan-related judgments regardless of which mallet was used. For the experiment described here, we chose *perceived hardness of mallet* as the perceptual dimension to investigate.

All pans were cast aluminum with copper bottoms. In each case, the pan was struck on the bottom between the center and the rim. Table 1 lists the sizes and shapes of pans, in order of size, and the types of mallets used, roughly in order of actual physical hardness.

Table 1

Pans
1-quart saucepan
2-quart saucepan
3-quart saucepan
6-quart dutch oven
Mallets
metal (spoon)
wood
rubber
cloth-covered wood
felt
felt-covered rubber

All recordings were made digitally using a Shure SM81 condenser microphone, a Sony PCM-F1 digital audio processor, a Sony SL-2005 portable videocassette recorder (Betamax), and a Sony Dycamion L-750UIG video cassette. The digital recordings were transferred directly to the computer using a digital interface for the PCM-F1 designed and built at NCM by Jim Roth. The sampling rate for all recordings was 44.1 kHz.

3.2 Perceptual ratings

For this pilot study, the two authors served as the only subjects. Each subject gave eight ratings of perceived mallet hardness for each of the 24 stimuli. A scale of 1 to 9 was used, with higher numbers indicating harder mallets. The average rating for each stimulus is plotted in Fig. 1 for one subject (WLM), along with standard deviation bars to indicate the variability of the ratings. Vertical dotted lines separate the results for the four different pans. The order of the regions is the same as the order of the pans in Table 1. Within each region, the results for the six different mallets are ordered as shown in Table 1.

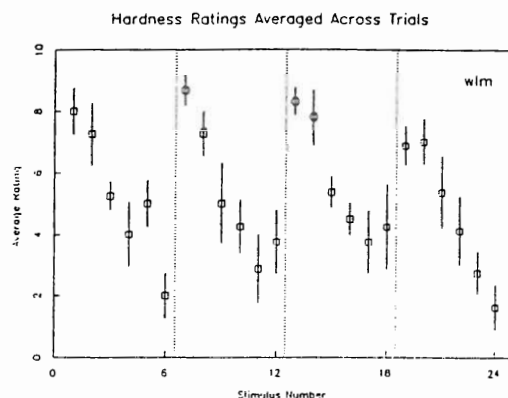


Fig. 1. Summary of responses for subject WLM.

One additional experiment was performed. We conjectured that the acoustical information which specifies properties of the mallet is restricted to the attack portion of the sound. As a preliminary step, we tried removing varying amounts of the initial portion of each sound to see at what point it became impossible to tell the mallets apart. Informal tests showed that the mallet-specific information is contained within the first 300 msec. This was formally tested by removing 300 msec from the beginning of each sound and collecting perceived mallet hardness judgments. The ratings verified the conjecture: hardness ratings were unrelated to the actual hardness of the mallet, and subjects were generally unable to hear *any* variation in mallet hardness across all the stimuli. This allowed us to consider the 300 msec point as a division between the attack and the decay of the stimuli in our subsequent acoustical analyses (see Sec. 3.3).

3.3 Acoustical parameters

We adopted an empirical approach to devising acoustical parameters. Preliminary analyses of the recordings indicated acoustical features which were likely physical correlates of mallet hardness; a total of 8 parameters were then designed to capture these features. The parameters are listed in Table 2, along with abbreviations for each that are used throughout the remainder of this paper.

For an excellent review of the acoustics of percussion instruments, including both theoretical models and empirical measurements, we refer the reader to *Percussive*

Table 2

Abbreviation	Acoustical Parameter	Average	Std. Dev.
ce.ratio	Spectral Centroid Log Ratio	0.4777	0.261
env.exp	Envelope Decay Constant	-4.876	6.528
env.peak	Envelope Amplitude Peak	0.0626	0.0013
envdev.peak	Envelope Deviation Peak	0.0703	0.0034
envdev.avg	Envelope Deviation Average	0.00199	0.000019
inharm.avg	Average Inharmonicity Ratio	0.6512	0.1051
rms	RMS Level	0.0258	0.0001
pitch	Pitch Match	370.91	10644.31

Notes, 19(3), Fall 1982 - a special issue devoted entirely to percussion acoustics (T. D. Rossing contributed the bulk of the material for the issue). In particular, Bassett (1982, same issue) discusses acoustical consequences of mallet hardness and size.

Preliminary measurements of spectral centroids of the stimuli showed that, for a given pan, the centroid remains fairly constant after the first 200 to 300 msec, and tends to be roughly equal for all mallets. In the attack portion, however, the centroid depends strongly on

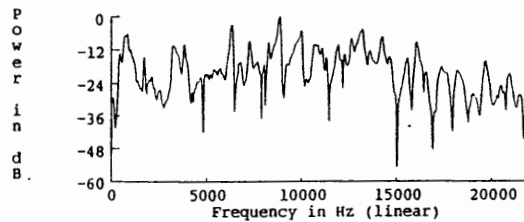


Fig. 2. Spectra for metal mallet, 1-quart saucepan. Top is attack spectrum (first 23 msec), bottom is decay spectrum (average spectrum after 300 msec).

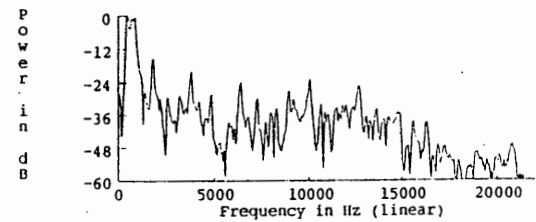


Fig. 3. Spectra for felt mallet, 1-quart saucepan. Top is attack spectrum (first 23 msec), bottom is decay spectrum (average spectrum after 300 msec).

3.3.1. Spectral distribution. Preliminary analyses indicated that spectral distribution is a significant cue to hardness of mallet. Figs. 2 and 3 show spectra of a hard mallet and a soft mallet, respectively, on the same pan. Each figure shows the attack spectrum (top half) and a longtime-average spectrum of the remainder of the sound (bottom half). It is clear that harder mallets contain more high-frequency energy, but this trend is only evident in the attack portion of the sound. After the attack, the sounds have roughly equivalent spectra.

Several single-valued measures of spectral distribution have been proposed in the literature (von Bismarck 1974b; Grey & Gordon 1978). We chose to follow Grey and Gordon (1978) by measuring *spectral centroid*. Given a sequence of spectral amplitudes A_i , $0 \leq i \leq N-1$, where each amplitude A_i corresponds to a frequency f_i , the spectral centroid may be defined as

$$\frac{\sum_{i=0}^{N-1} A_i f_i}{\sum_{i=0}^{N-1} A_i}$$

the mallet (see Fig. 4). For hard mallets, the centroid begins well above the final value, and decays rapidly; for soft mallets, the initial centroid value is only slightly above the final value.

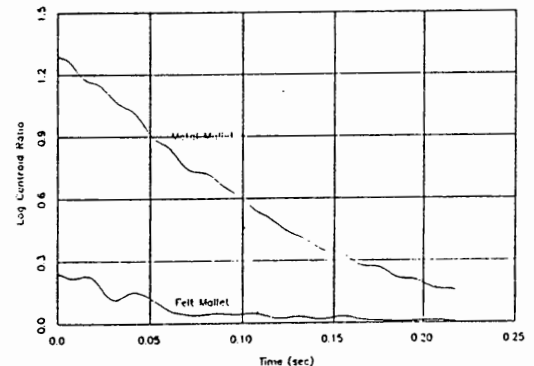
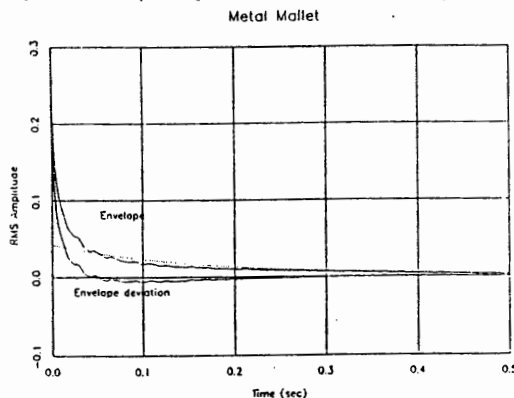


Fig. 4. Log of spectral centroid ratio as a function of time for metal and felt mallets, 1-quart saucepan. Centroids calculated every 1.5 msec.

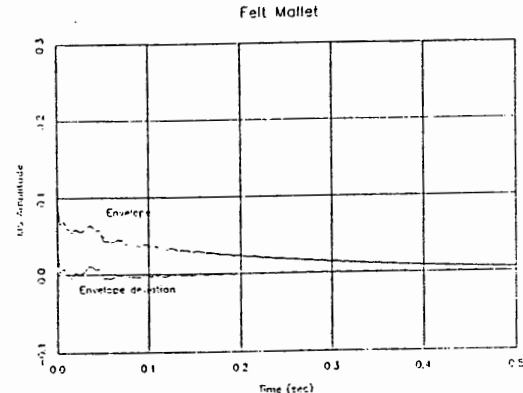
These measurements indicated that centroid values after the attack are a function of the pan, while centroid values during the attack are a function of the mallet. We therefore defined one acoustical parameter ("ce.ratio") to be $\log_{10} \frac{C_{att}}{C_{rem}}$, where C_{att} is the spectral centroid of the attack, and C_{rem} is the spectral centroid of the remaining portion of the sound. C_{att} was derived from a single FFT of the first 23 msec. C_{rem} was derived from a longtime average spectrum of the signal with the first 300 msec removed, calculated by averaging FFT's (23 msec windows with 50% overlap).

3.3.2. Envelope shape. Preliminary measurements of envelope (RMS amplitude over time) indicated that harder mallets are roughly characterized by a steeper attack slope followed by a rapid initial decay (Figs. 5 and 6). We used regression techniques to fit the envelope of each stimulus with a curve of the form Ae^{Bt} , where t is time. The coefficients A and B were included as acoustical parameters ("env.peak" and "env.exp," respectively).



As Moorer describes it, "the method consists of taking a small window and determining the [nonrecursive] comb filter which when applied to the input sequence represented by this window produced the minimum output...." (1974, p. 330). This filter is the optimum comb filter, and the reciprocal of the optimum comb filter delay is taken as the fundamental frequency estimate. The algorithm requires that a finite range of expected fundamental frequency values be specified, so that a finite number of comb delays may be tested. To determine this range, we used an alternate pitch tracking algorithm, devised by Rabiner (Rabiner and Schafer 1978, pp. 135-141) and implemented by Mark Dolson of CARL (the Computer Audio Research Laboratory of the University of California, San Diego, USA).

To see how the algorithm provides a measure of inharmonicity, it must be recognized that the energy passed by each trial comb filter functions as a measure of goodness of fit. The more closely the comb delay matches the pitch period, the less energy will be passed by the comb filter. The main characteristic of inharmonic signals is



Figs. 5 and 6. Envelope curves for metal and felt mallets, 1-quart saucepan. Dotted lines are exponential curves fit to the actual envelope. Amplitudes calculated every 2.5 msec.

We then subtracted the fit curve from the actual envelope to obtain the *envelope deviation* curve. Figs. 5 and 6 show the envelope, fit curve, and envelope deviation for a hard mallet and a soft mallet, respectively. The steep attack slope of the hard mallet is represented as a sharp initial peak in the envelope deviation. The envelope deviation of the soft mallet lacks this feature. Consequently we defined two new acoustical parameters: the average of the first 300 msec of the envelope deviation curve ("envdev.avg"), and the peak positive value of the envelope deviation ("envdev.peak").

3.3.3. Inharmonicity. A prominent feature of "metallic-sounding" timbres in general is inharmonicity, which may be defined as the deviation of significant frequency components from a harmonic relationship. Quantifying the extent of this deviation requires two steps. First, the best estimate of the fundamental frequency must be obtained. Second, the deviation of the components from multiples of the fundamental frequency must be measured. Moorer's (1974) *optimum comb method*, originally intended only as a means of estimating fundamental frequency, is also useful for providing a measure of inharmonicity.

that they have no clear fundamental frequency. It can therefore be expected that, for inharmonic inputs, a large amount of energy will be passed by each trial comb filter, including the optimum comb. We defined the *inharmonicity ratio* to be $\frac{E_{in}}{E_{comb}}$, where E_{in} is the energy of the input segment and E_{comb} is the energy of the output of the optimum comb for that segment (following Moorer's example, a sum of absolute values was used as a measure of energy rather than the usual RMS for increased efficiency).

Figs. 7 and 8 show plots of inharmonicity ratio over time for hard and soft mallets for two different pans. Three trends are apparent. First, inharmonicity ratios are consistently higher for the 2-quart saucepan than for the 1-quart saucepan. In informal listening sessions, we judged the 2-quart saucepan sounds to be more "clangorous" and to have a less definite sense of pitch than the 1-quart saucepan sounds; we take this as evidence that our measure of inharmonicity has some validity. Second, for both pans, inharmonicity is greatest in the attack and decreases throughout the decay; this also corresponds with informal evaluations. Third, for the 1-quart pan

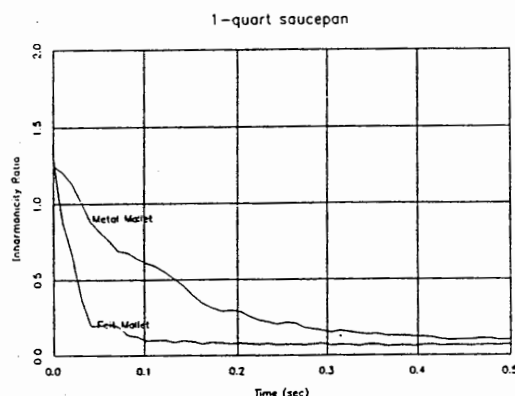


Fig. 7. Inharmonicity ratio as a function of time for metal and felt mallets, 1-quart saucepan. Values calculated every 10 msec.

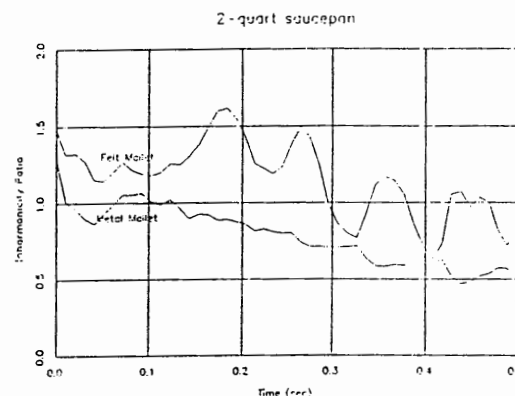


Fig. 8. Inharmonicity ratio as a function of time for metal and felt mallets, 2-quart saucepan. Values calculated every 10 msec.

only, there is a mallet-dependent trend: inharmonicity decreases rapidly during the attack for soft mallets and slowly for hard mallets. We defined another acoustical parameter ("inharm.avg") to be the average value of the inharmonicity ratio over the first 300 msec.

3.3.4. Other parameters. We defined two additional acoustical parameters which we did not expect to play major roles: the overall RMS of the stimulus ("rms"), and the average fundamental frequency value ("pitch"), which was calculated using Rabiner's method (see Sec. 3.3.3).

3.4 Analysis of acoustical parameters

All 21 of our stimuli were measured on the 8 acoustical parameters described in Sec. 3.3, and these data were submitted to a correlation analysis. Table 3 shows the resultant correlation matrix. In order to give a better idea of how the values on each parameter interrelate for our stimuli, scatter plots of all pair-wise comparisons of the acoustical measurements are presented in Fig. 9.

Two instances of high correlations are immediately apparent in the scatter plots, and can be confirmed by referring to the correlation matrix in Table 3. First, envdev.peak may be seen to increase with ce.ratio, and the matrix shows a high positive correlation of 0.6529. This correlation is not surprising; the high-frequency energy present in the attack of signals with high ce.ratio values should elevate the attack portion of the envelope

accordingly. Second, the highest correlation, 0.8582, was found between rms and env.peak, since both measures are related to overall energy level. A strong negative correlation, -0.6477, was obtained between envdev.avg and env.peak; apparently, those stimuli with a sharp initial attack drop more quickly in amplitude than those with a more gradual attack and undershoot the overall exponential shape of the envelope.

The acoustical measures were also submitted to a principal components analysis in order to determine whether a smaller number of acoustical predictors could adequately describe the variation in the stimuli. For our eight acoustical parameters, principal components analysis found eight orthogonal vectors in an eight-dimensional space that are linear combinations (*rotations*) of the eight original parameters. The first component was chosen to include the maximum amount of variance in the data, and each successive component accounts for as much of the remaining variance as possible. Before the analysis was performed, the measurements on each parameter were rescaled ("centered") by subtracting the mean on each parameter from all the measurements on that parameter and then dividing by the standard deviation on that parameter (see Table 2). This yielded a set of acoustical measures which all have a mean of 0.0 and a standard deviation of 1.0. The rotation of the original acoustical parameters resulted in a redistribution of the variance illustrated in Fig. 10 by the standard deviation for each of the derived principal components.

Table 3

Correlation Matrix for Acoustical Parameters							
Parameter	ce.ratio	env.exp	env.peak	envdev.peak	envdev.avg	inharm.avg	rms
env.exp	0.2350						
env.peak	-0.3193	0.0655					
envdev.peak	0.7877	-0.0162	-0.2365				
envdev.avg	0.4684	-0.1468	-0.6477	0.6529			
inharm.avg	-0.1482	-0.5614	-0.1747	0.1567	0.5023		
rms	-0.1107	-0.1541	0.8582	0.1397	-0.2748	-0.0063	
pitch	-0.1493	-0.7379	0.0448	-0.1695	-0.2535	0.0009	0.1155

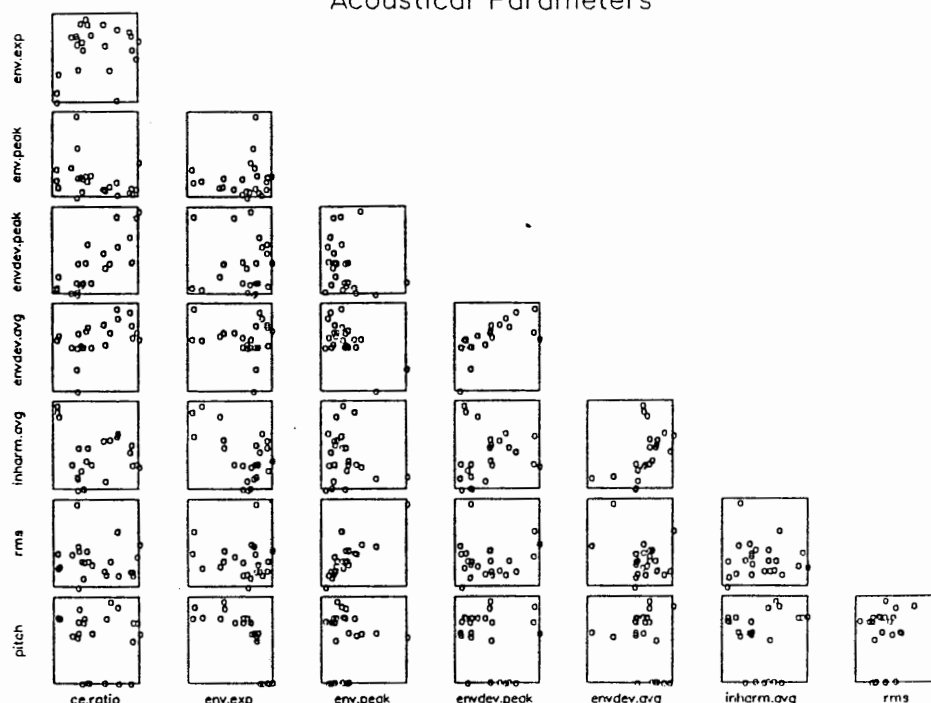


Fig. 9. Scatter plots for all pair-wise comparisons of acoustical parameters.

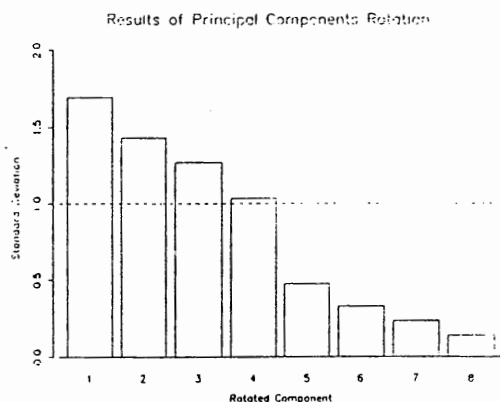


Fig. 10. Distribution of variance among principal components.

Four of the components account for more of the variance than any of the original measures, but only the first two components are easily identifiable. The first component puts a high weight on *ce.ratio* and envelope factors (positive on *envdev.avg* and *envdev.peak*, negative on *env.peak*), and seems to contain variance due to mallet effects. The second component puts a high weight on *pitch*, *env.exp*, and *inharm.avg*. With just a few exceptions, these three parameters vary primarily from pan to

pan. For example, *pitch* varies considerably from pan to pan, but not at all among different mallets on the same pan. Table 4 shows the linear weights applied to values on each of the original measures that define the rotation to the principal components.

3.5 Derivation of the psychophysical relation

Since information on mallet hardness seems to be carried by several acoustical cues, we chose to derive a many-to-one rather than a one-to-one psychophysical relation. Eight subjective ratings of perceived hardness of mallet for each stimulus and from two listeners were combined (192 datapoints) for a multiple regression analysis using our eight acoustical measures as predictors. As for the principal components analysis, the acoustical measures were all rescaled to mean 0.0 and standard deviation 1.0. Consequently, the regression coefficients found in Table 5 give a clear picture of how good a predictor of perceived hardness each acoustical measure is (the subjective ratings were rescaled in the same manner for the analysis).

The regression equation described by the coefficients in Table 5 accounts for 63.94 % of the variance in the hardness ratings. The strongest predictors of perceived hardness of the eight parameters are *pitch*, *ce.ratio*, and *inharm.avg*. All three predictors make a significant contribution to the fit, with *t* values exceeding the criterion value of 3.373 for $p < .001$.

Table 4

Orthogonal Matrix for Principal Components Rotation								
Parameter	Component Number							
	1	2	3	4	5	6	7	8
ce.ratio	0.4131	-0.2052	0.3210	-0.4048	0.5245	-0.3904	0.2934	-0.0806
env.exp	0.0352	-0.6777	-0.0759	0.1315	0.0688	-0.1913	-0.6749	-0.1392
env.peak	-0.4632	-0.1006	0.4429	0.1583	0.1978	-0.1379	-0.0281	0.7036
envdev.peak	0.4257	-0.0188	0.5026	-0.1303	-0.0796	0.6730	-0.2674	0.1341
envdev.avg	0.5265	0.1811	0.0863	0.2084	-0.5306	-0.5091	-0.0780	0.3145
inharm.avg	0.1779	0.4857	0.0582	0.5707	0.5505	-0.0449	-0.2733	-0.1528
rms	-0.2976	0.0436	0.6568	0.1231	-0.2998	-0.1839	0.0322	-0.5814
pitch	-0.1863	0.4735	-0.0051	-0.6264	0.0081	-0.2092	-0.5514	0.0239

Table 5

Multiple Regression Results				
No.	Parameter	Coef	Std Err	t Value
1	ce.ratio	0.6199541	0.0998097	6.211364
2	env.exp	0.4061472	0.1428319	2.843532
3	env.peak	0.2813567	0.2318433	1.213564
4	envdev.peak	0.2156532	0.1175110	1.835174
5	envdev.avg	0.0797341	0.1363172	0.584916
6	inharm.avg	0.4223991	0.0947328	4.458846
7	rms	-0.0926540	0.1950096	-0.475125
8	pitch	0.6332255	0.1153565	5.489292

All Possible Subsets of Predictors

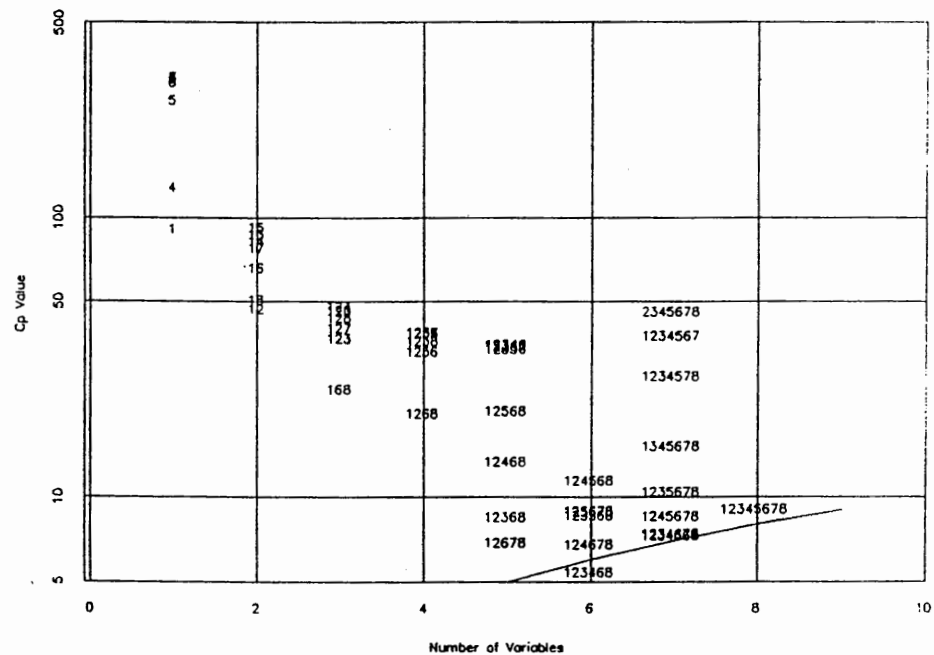


Fig. 11. C_p values for all possible subsets of predictors. Smaller C_p values indicate a better fit. The line near the bottom of the graph shows the expected value for a good fit.

Contour Plot of Average Log Hardness

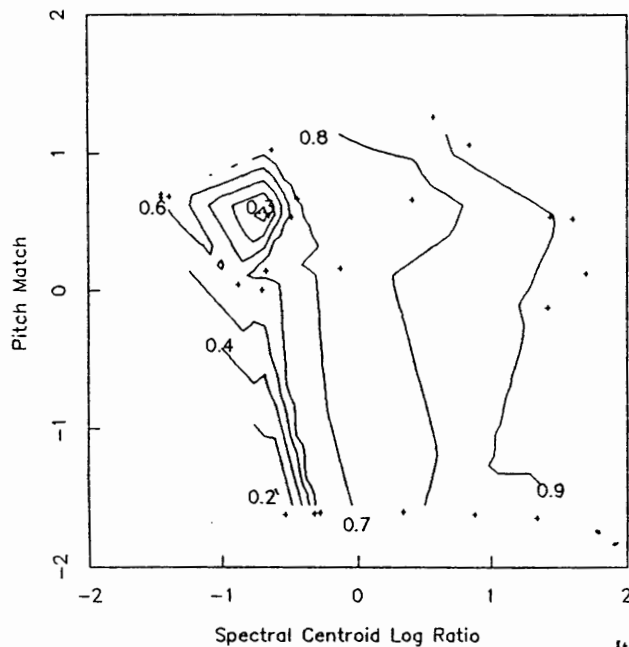


Fig. 12. Average log hardness rating as a function of two predictors, pitch and ce.ratio, shown in a contour plot. The "+" symbols indicate the locations of data points.

Since these three predictors are stronger than any others, it would be interesting to know whether the three predictors, taken alone, would yield as good a fit to the hardness ratings as did the full regression. We can evaluate all possible subsets of predictors for our regression analysis using the C_p statistic (Mallows 1973). Fig. 11 plots C_p values for all possible subsets of our predictors as a function of the number of variables included in the regression analysis. Smaller C_p values indicate a better fit. The line near the bottom of Fig. 11 shows the expected value for a good fit.

The symbols used to plot C_p value name the variables included in the regression analysis according to the numbering of our acoustical measures from one to eight (listed in Table 5). Notice that no one predictor does very well by itself. Predictors 1, 4, and 5 (ce.ratio, envdev.peak, and envdev.avg) do best of the eight predictors, but have very high C_p values. Of the two-variable subsets, 1 combined with 2 does best (ce.ratio and env.peak). The best fit with three variables is 1, 6, and 8 (ce.ratio, inharm.avg, and pitch). Only when six variables are included in the regression does the value of C_p become small enough to indicate a good fit.

Perspective Plot of Average Log Hardness

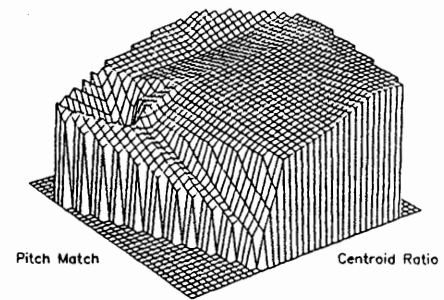


Fig. 13. Average log hardness rating as a function of two predictors, pitch and ce.ratio, shown in a perspective plot.

It remains to be seen whether any of these regression equations can predict what hardness of mallet judgment would result for an arbitrary stimulus. In order to make such a prediction, one would first measure the stimulus on all eight acoustical parameters and sum up the eight terms given by the product of each measurement and the coefficient for the corresponding parameter. If, however, the stimulus set upon which the regression coefficients are based did not cover the range of sounds for which predictions are desired, it is possible that extrapolation to new stimuli would incur large errors.

For example, even if the coefficients for pitch and ce.ratio are based upon an idiosyncratic stimulus set, we can still fit a smooth regression equation to capture how hardness judgments will change with these predictors. Fig. 12 shows a contour plot of hardness rating as a function of pitch and ce.ratio. The "+" symbols indicate the locations of the actual data points upon which the contour plot is based. The perspective plot of the same surface in Fig. 13 shows there is no doubt that for our stimulus set, hardness judgments increased both with pitch and ce.ratio. A different sampling of pitch values within a stimulus set would probably put a different weight upon pitch as a predictor of hardness judgments. This is always a potential problem for arbitrarily selected stimuli. This pitfall can be circumvented for perceptual control of synthesis through an iterative procedure of psychophysics and synthesis. The procedure calls for selecting stimuli on the basis of psychophysical results, collecting and analyzing psychophysical data on the stimulus set, and then selecting stimuli on the basis of the new results. A few iterations should allow the stimulus set to sample broadly and evenly the range of timbres of interest.

4. CONCLUSION

The initial goal of the pilot study – to derive a psychophysical relation for predicting perceived hardness of mallet judgments – has been achieved. Our experimental paradigm has proven successful at indicating which acoustical parameters are relevant. The unexpected discovery that pitch affects mallet hardness judgments illustrates the importance of evaluating statistically the influence of acoustical properties as opposed to making *a priori* decisions.

The next phase of the research will be to verify and refine our psychophysical relation through the iterative strategy described in Sec. 3.5. Synthesis of new stimuli also will allow us to test the relative importance of redundant cues. Acoustical parameters found to be highly correlated in the original stimuli will be set in conflict with each other in the synthesized stimuli. For example, one parameter will be set to produce a high mallet hardness judgment, and another parameter will be set to produce a low mallet hardness judgment. The actual rating received by this new stimulus will indicate which parameter has the dominant effect and what happens outside of the typical range of pairings of acoustical values.

The ability to predict judgments about perceived mallet hardness may seem so specialized as to be of little practical value for musicians. We feel, however, that a wide variety of timbres have attack-related qualities, such as "edge" or "bite," which are analogous to mallet hardness. We predict that an understanding of the psychophysics of mallet hardness will be an excellent foundation for more general research on the timbral effects of attack transients.

ACKNOWLEDGMENTS

Our thanks go to the entire staff at Northwestern Computer Music (NCM) for support of all sorts – software, hardware, and camaraderie. Special thanks are due to Gary Kendall, for numerous helpful suggestions, and to Jim Roth, Doug Jones, and Nancy Barker Walczak, who assisted in the recordings. We are also indebted to our colleagues at CARL and IRCAM for providing various indispensable bits of software. This research would not have been possible without the support provided to NCM by a grant from the System Development Foundation.

REFERENCES

- Adrien, J.-M., & Rodet, X. (1985) "Physical Models of Instruments: A Modular Approach, Application to Strings." *Proc. of the Int'l Computer Music Conf. 1985*, ed. B. Truax, San Francisco: Computer Music Association, 85–89.
- Bassett, I. G. (1982) "Vibration and Sound of the Bass Drum." *Percussive Notes*, 19(3), 50–58.
- von Bismarck, G. (1974a) "Timbre of Steady Sounds: A Factorial Investigation of Its Verbal Attributes." *Acustica*, 30, 146–150.
- von Bismarck, G. (1974b) "Sharpness as an Attribute of the Timbre of Steady Sounds." *Acustica*, 30, 159–172.
- de Bruijn, A. (1978) "Timbre-Classification of Complex Tones." *Acustica*, 40, 108–114.
- Cadoz, C., Luciani, A., & Florens, J. (1984) "Responsive Input Devices and Sound Synthesis by Simulation of Instrumental Mechanisms: The Cordis System." *Computer Music J.*, 8(3), 60–73.
- Cermak, G. W., & Cornillon, P. C. (1976) "Multidimensional Analyses of Judgments About Traffic Noise." *J. Acoust. Soc. Amer.*, 59(6), 1412–1420.
- Charbonneau, G. R. (1981) "Timbre and the Perceptual Effects of Three Types of Data Reduction." *Computer Music J.*, 5(2), 10–19.
- Chowning, J. M. (1980) "Computer Synthesis of the Singing Voice." In J. Sundberg (Ed.), *Sound Generation in Winds, Strings, Computers*, Stockholm: Royal Swedish Academy, 4–13.
- Fletcher, H., & Bassett, I. G. (1978) "Some Experiments with the Bass Drum." *J. Acoust. Soc. Amer.*, 64(6), 1570–1576.
- Gibson, J. J. (1966) *The Senses Considered as Perceptual Systems*. Boston: Houghton Mifflin.
- Grey, J. M. (1975) "An Exploration of Musical Timbre." Ph. D. diss., Department of Music, Stanford University, Stanford, CA, USA. Available as Department of Music Report No. STAN-M-2.
- Grey, J. M. (1977) "Multidimensional Perceptual Scaling of Musical Timbres." *J. Acoust. Soc. Amer.*, 61(5), 1270–1277.
- Grey, J. M., & Gordon, J. W. (1978) "Perceptual Effects of Spectral Modifications on Musical Timbres." *J. Acoust. Soc. Amer.*, 63(5), 1493–1500.
- Grey, J. M., & Moorer, J. A. (1977) "Perceptual Evaluations of Synthesized Musical Instrument Tones." *J. Acoust. Soc. Amer.*, 62(2), 454–462.
- Hiller, L., & Ruiz, P. (1971) "Synthesizing Musical Sounds by Solving the Wave Equation for Vibrating Objects – Part I." *J. Audio Eng. Soc.*, 19(6), 462–470.
- Howard, J. H., Jr., & Silverman, E. B. (1976) "A Multidimensional Scaling Analysis of 16 Complex Sounds." *Perception and Psychophysics*, 19(2), 193–200.
- Luce, D., & Clark, M., Jr. (1967) "Physical Correlates of Brass-Instrument Tones." *J. Acoust. Soc. Amer.*, 42(6), 1232–1243.
- Mallows, C. L. (1973) "Some comments on Cp." *Technometrics*, 15, 661–675.
- Martens, W. L. (1985) "Palette: An Environment for Developing an Individualized Set of Psychophysically Scaled Timbres." *Proc. of the Int'l Computer Music Conf. 1985*, ed. B. Truax, San Francisco: Computer Music Association, 355–365.

- McAdams, S. (1982) "Spectral Fusion and the Creation of Auditory Images." In M. Clynes (Ed.), *Music, Mind, and Brain: The Neuropsychology of Music*. New York: Plenum, 279-298.
- McAdams, S., & Saarialho, K. (1985) "Qualities and Functions of Musical Timbre." *Proc. of the Int'l Computer Music Conf. 1985*, ed. B. Truax, San Francisco: Computer Music Association, 367-374.
- McNabb, M. (1981) "Dreamsong: The Composition." *Computer Music J.*, 5(4), 36-53.
- Miller, J. R., & Carterette, E. C. (1975) "Perceptual Space for Musical Structures." *J. Acoust. Soc. Amer.*, 58(3), 711-720.
- Moorer, J. A. (1974) "The Optimum Comb Method of Pitch Period Analysis of Continuous Digitized Speech." *IEEE Trans. on Acoust., Speech, and Sig. Proc.*, ASSP-22(5), 330-338.
- Neisser, U. (1976) *Cognition and Reality*. San Francisco: Freeman.
- Plomp, R. (1970) "Timbre as a Multidimensional Attribute of Complex Tones." In R. Plomp & G. F. Smoorenburg (Eds.), *Frequency Analysis and Periodicity Detection in Hearing*, Leiden: A. W. Sijthoff, 397-414.
- Plomp, R., Pols, L. C. W., & van de Geer, J. P. (1967) "Dimensional Analysis of Vowel Spectra." *J. Acoust. Soc. Amer.*, 41(3), 707-712.
- Plomp, R., & Steencken, H. J. M. (1969) "Effect of Phase on the Timbre of Complex Tones." *J. Acoust. Soc. Amer.*, 46(No. 2, Part 2), 409-421.
- Rabiner, L. R., & Schafer, R. W. (1978) *Digital Processing of Speech Signals*. Englewood Cliffs, NJ: Prentice-Hall.
- Risset, J.-C., & Mathews, M. V. (1969) "Analysis of Musical-Instrument Tones." *Physics Today*, 22(2), 23-30.
- Rossing, T. D., & Sykes, W. A. (1982) "Acoustics of Indian Drums." *Percussive Notes*, 19(3), 58-67.
- Searle, C. L. (1982) "Speech Perception from an Auditory and Visual Viewpoint." *Canadian J. Psychology*, 36(3), 402-419.
- Solomon, L. N. (1958) "Semantic Approach to the Perception of Complex Sounds." *J. Acoust. Soc. Amer.*, 30(5), 421-425.
- Studdert-Kennedy, M. (1985) "Perceiving Phonetic Events." In W. H. Warren, Jr., & R. E. Shaw (Eds.), *Persistence and Change: Proceedings of the First International Conference on Event Perception*, Hillsdale, NJ: Erlbaum, 139-156.
- Terhardt, E. (1974) "On the Perception of Periodic Sound Fluctuations (Roughness)." *Acustica*, 30, 201-213.
- VanDerveer, N. J. (1979) "Ecological Acoustics: Human Perception of Environmental Sounds." Ph. D. diss., Department of Psychology, Cornell University, Ithaca, NY, USA.
- Warren, W. H., Jr., & Verbrugge, R. R. (1984) "Auditory Perception of Breaking and Bouncing Events: A Case Study in Ecological Acoustics." *J. Experimental Psychology - Human Perception and Performance*, 10(5), 704-712.
- Wedin, L., & Goude, G. (1972) "Dimension Analysis and the Perception of Instrumental Timbre." *Scandinavian J. Psychology*, 13(3), 228-240.
- Wessel, D. L. (1979) "Timbre Space as a Musical Control Structure." *Computer Music J.*, 3(2), 45-52.
- Zahorian, S. A., & Rothenberg, M. (1981) "Principal-Components Analysis for Low-Redundancy Encoding of Speech Spectra." *J. Acoust. Soc. Amer.*, 69(3), 832-845.